Chapter 1, Problem 24P

Problem

A utility company charges 8.2 cents/kWh. If a consumer operates a 60-W light bulb continuously for one day, how much is the consumer charged?

Step-by-step solution

Step 1 of 3

The wattage of the bulb $p = 60\text{ W}$

The time that the bulb operates $t = 1\,\text{day}$

The charges for $1\,\text{kWh} = 8.2\,\text{cents}$

Step 2 of 3

Calculate the energy consumed by the bulb for one day.

$$\begin{aligned} w &= pt \\ &= (60\text{ W})(24\text{ h}) \\ &= 1440\text{ Wh} \\ &= 1.44\text{ kWh} \end{aligned}$$

Step 3 of 3

Calculate the consumer charge for 1.44kWh energy.

$$\begin{aligned} \text{consumer charge} &= 1.44 \times 8.2\,\text{cents} \\ &= 11.80\,\text{cents} \end{aligned}$$

Therefore, the consumer charged for one day is $\boxed{11.80\,\text{cents}}$.